Rishi D. **Jha**

PhD Candidate · AI Security & Safety

rjha@cs.cornell.edu | rishijha.com | rjha18 | rishi-jha | rishi_d_jha

Education

Cornell Tech — *New York / Ithaca, NY*

PhD Candidate, Computer Science — *Aug 2023 – Present*

- Advisor: Prof. Vitaly Shmatikov

University of Washington — Seattle — *Seattle, WA*

MS., Computer Science — *Sep 2022 – Jun 2023*

- Master's Thesis: *Label Poisoning is All You Need*
- Advisor: Prof. Sewoong Oh

University of Washington — Seattle — *Seattle, WA*

BS.BA., Computer Science and Mathematics — Philosophy: *Cum Laude, Phi Beta Kappa* — *Sep 2018 – Mar 2022*

- Jun 2022: Graduated Cum Laude with Phi Beta Kappa honors
- 2018–2022: Dean's List (all eligible quarters)

Awards and Honors

2026	**Spotlight Talk**, MATS Symposium — Top 10% of fellows	*Berkeley, CA*
2026	**Fellow**, MATS — 5% of applicants	*Berkeley, CA*
2025	**Doctoral Fellow**, Digital Life Initiative	*New York, NY*
2024	**Distinguished Paper Award**, USENIX Security — Top 22 papers (out of 417)	*Philadelphia, PA*
2024	**Graduate Research Fellowship Program Honorable Mention**, NSF	*USA*
2023	**Cornell University Fellow**, Cornell University — 20% of incoming PhDs	*Ithaca, NY*
2022	**Phi Beta Kappa**, University of Washington	*Seattle, WA*
2022	**Cum Laude**, University of Washington — Top 10% across Arts & Sciences	*Seattle, WA*
2018–22	**Dean's List**, University of Washington — All eligible quarters	*Seattle, WA*
2021–22	**Varsity Climbing Team**, University of Washington	*Seattle, WA*

Publications

Conference

[1] **Agent Meltdowns: The Road to Hell Is Paved with Helpful Agents**. **Submission to NeurIPS 2026**.
Rishi Jha*, Harold Triedman*, Arkaprabha Bhattacharya, and Vitaly Shmatikov.

[2] **Breaking and Fixing Defenses Against Control Flow Hijacking in Multi-Agent Systems**. **ICLR 2026**.
Rishi Jha, Harold Triedman, Justin Wagle, and Vitaly Shmatikov.

[3] **Adversarial Hubness in Multi-Modal Retrieval**. **IEEE S&P 2026**.
Tingwei Zhang, Fnu Suya, Rishi Jha, Collin Zhang, and Vitaly Shmatikov.

[4] **Platonic Representations in the Human Brain: Unsupervised Recovery of Universal Geometry**. **Submission to NeurIPS 2026**.
Pablo Marcos-Manchón, Rishi Jha, and Lluís Fuentemilla.

[5] **Harnessing the Universal Geometry of Embeddings**. **NeurIPS 2025**.
Rishi Jha, Collin Zhang, Vitaly Shmatikov, and John X. Morris.

[6] **Multi-Agent Systems Execute Arbitrary Malicious Code**. **COLM 2025**.
Harold Triedman, Rishi Jha, and Vitaly Shmatikov.

[7] **Adversarial Illusions in Multi-Modal Embeddings**. **USENIX Security 2024**. 🏆 Distinguished Paper Award; Top 5%.
Tingwei Zhang*, Rishi Jha*, Eugene Bagdasaryan, and Vitaly Shmatikov.

[8] **Label Poisoning is All You Need**. **NeurIPS 2023**.
Rishi Jha*, Jonathan Hayase*, and Sewoong Oh.

[9] **Hyper-Universal Policy Approximation: Learning to Generate Actions from a Single Image using Hypernets**. **Neurovision @ CVPR 2022**.
Dimitrios C. Gklezakos, Rishi Jha, and Rajesh P.N. Rao.

[10] **On Geodesic Distances and Contextual Embedding Compression for Text Classification**. **TextGraphs @ NAACL 2021**.
Rishi Jha and Kai Mihata.

Master's Thesis

[11] **Label Poisoning is All You Need**. **University of Washington**. **2023**.

Patents

[12] **Graph-Based Analysis of Security Incidents**. **U.S. Patent 12081569**. **2024**.
Nisha S. Hameed, Rishi D. Jha, and Evan Argyle.

Invited Talks & Guest Lectures

Invited

2026	**Google**, "The Road to Hell Is Paved with Persistent Agents." *Adversarial Knowledge Share Series*.	*Remote*
2026	**MATS**, "The Road to Hell Is Paved with Persistent Agents." *MATS Spotlight Talk*.	*Berkeley, CA*
2026	**Cornell Tech**, "All AI Models Might be the Same: Harnessing..." *Digital Life Initiative*.	*New York, NY*
2026	**Harvard**, "All AI Models Might be the Same: Harnessing..." *AstroAI Colloquium*.	*Cambridge, MA*
2026	**MIT**, "All AI Models Might be the Same: Harnessing..." *Platonic Space Meeting*.	*Cambridge, MA*
2025	**Tufts**, "All AI Models Might be the Same: Harnessing..." *Computational Platonic Space*.	*Remote*
2025	**UC Berkeley**, "Breaking and Fixing Multi-Agent Systems." *Security & AI Seminar*.	*Berkeley, CA*
2025	**Microsoft ASG**, "Contextual Control and Information Flow in Multi-Agent Systems."	*Redmond, WA*
2025	**Planet Labs**, "Harnessing the Universal Geometry of Embeddings."	*Remote*
2025	**MSR**, "Multi-Agent Systems Execute Arbitrary Malicious Code." *Security & Privacy Workshop*.	*Redmond, WA*
2025	**MongoDB**, "Multi-Agent Systems Execute Arbitrary Malicious Code."	*New York, NY*
2025	**Microsoft ASG**, "Emergent Risks in Multimodality."	*Remote*

Guest Lectures

2025	**Cornell Tech**, "Hacking AI Agents."	*New York, NY*
2025	**Cornell Tech**, "Security & Privacy in Machine Learning."	*New York, NY*
2021	**The Overlake School**, "Topics on Sparse Coding."	*Remote*

Academic Research

Cornell Security — *New York, NY*

PhD Candidate — *Aug 2023 – Present*

Working with **Prof. Vitaly Shmatikov** on agentic ([1, 2, 6]) and embedding ([5, 4, 3, 7]) security and safety. Selected projects:

- Discovered *agent meltdowns* in which benign task failures drive frontier agents into overpersistent recovery and dangerous behaviors like privilege escalation and doxxing. We developed a 13-behavior taxonomy and found medium/high-severity meltdowns in nearly 65% of blocking-error rollouts. **Submitted to NeurIPS 2026** [1].
- Introduced `vec2vec`, the first unsupervised method for unpaired cross-model embedding translation—showing encoders share a *universal geometry* across paradigms and vintages; enables cross-model interpretability and reveals sensitive-data leakage risks in vector databases. **NeurIPS 2025** [5].
- Discovered *control-flow hijacking* attacks in LLM-based multi-agent systems that exploit inter-agent communication to trigger arbitrary code execution and data exfiltration, achieving 45–100% success rates. **CoLM 2025** [6].
- Demonstrated *adversarial illusions* in commercial multimodal embeddings—small perturbations align an image/audio embedding to an arbitrary cross-modal target, enabling task-agnostic attacks. **Distinguished Paper Award, USENIX Security 2024** [7].

Sewoong Lab — Foundations of Machine Learning — *Seattle, WA*

Graduate Research Assistant — *May 2021 – Aug 2023*

Worked with **Prof. Sewoong Oh** and **Jonathan Hayase** on backdoor attacks and representation learning.

- Developed FLIP, a trajectory-matching backdoor attack that poisons only the training labels—leaving inputs untouched—to inject arbitrary triggers; achieves 99.6% success with <1% label corruption and <1% clean accuracy drop. Robust across datasets, triggers, and architectures. **NeurIPS 2023**, Master's thesis [8, 11].
- Created an open-source backdoor attack benchmark suite and survey: **backdoor-suite**.

Center for Neurotechnology — *Seattle, WA*

Undergraduate ML Researcher — *Mar 2020 – Aug 2022*

Worked with **Prof. Rajesh Rao** and **Dimitrios Gklezakos** on hypernetworks and neural representation learning.

- Co-developed HUPA, a personalized hypernetwork for task-conditional RL with up to 35% more resilience to sparsity and 25% better generalization than traditional embeddings. **NeuroVision @ CVPR 2022** [9].
- Built an audio-visual hypernetwork where video inputs control audio processing weights for large-scale cross-modal learning.
- Designed a convolutional manifold-learning network for unsupervised sparse feature learning in natural images.

Other Research — *Seattle, WA*

Undergraduate Student — *Nov 2018 – Jun 2021*

- (Self-Directed, Nov 2020 – Jun 2021) Worked with **Kai Mihata** to show that nonlinear compression of BERT embeddings can preserve downstream performance, making it practical for memory-constrained settings. **TextGraphs @ NAACL 2021** [10].
- (ICTD Lab, Nov 2018 – May 2019) Worked with **Spencer Sevilla** to evaluate chat app performance under poor network conditions and implement a teaching solution for schoolchildren in rural Indonesia.

Industry Research

MATS (with Google DeepMind) — *Berkeley, CA*

Fellow — *Jun 2026 – Present*

Working with **David Lindner** to develop low-cost mitigation methods for agentic overpersistence (as in [1]). We show that overpersistence begins in supervised finetuning, and cheap "lessons"-based SFT data can help mitigate. Submission in progress.

Google Research — *New York, NY*

Student Researcher — *Jan 2026 – Aug 2026*

Working with **Reza Shokri** to develop agentic memory defenses. We show that attacks on agentic memory can be framed as a "confused deputy" problem, and most modern memory methods are vulnerable to simple injections. We propose a control-flow defense that replaces existing methods that are circumvented by adaptive attackers. Submission in progress.

Microsoft — *Redmond, WA*

Research Intern — *May 2025 – Aug 2025*

Worked with **Justin Wagle** and **Kazuhito Koishida** on contextual multi-agent defenses.

- Demonstrated control-flow hijacking attacks evade alignment checks and goal drift detection in multi-agent systems due to brittle definitions and incomplete context visibility. Designed and implemented `ControlValve`, applying control-flow integrity and least privilege to automatically generate and enforce permitted control-flow graphs with zero-shot contextual rules. **ICLR 2026** [2].
- Project chosen as the core safety layer for upcoming deployments; internal conversations kicked off cross-team planning.

Microsoft Defender Research — *Redmond, WA*

Software Engineering Intern — Data Science — *Jul 2022 – Sep 2022*

- Ideated and implemented a low-cost, interpretable meta-learning framework that exploits spectral similarities in existing classifier responses to drive robustness in the Defender product. The system was lightweight, had upwards of 97% precision and recall.
- After my departure, the model was being pushed from pre-production to production for billions of users by the **end of 2023**.

Microsoft Defender Research — *Remote*

Software Engineering Intern — Data Science — *Jun 2021 – Sep 2021*

Patented in **September 2024** [12].

- Ideated and designed patented approach to detect malicious Command-and-Control intrusions in corporate networks using spectral methods on graphs. The model achieved high precision and recall in finding Indicators of Compromise in historical data.
- The project received significant investment from the team and Microsoft Research (MSR) since my departure with a goal of pushing an extension of the model to production in **Summer 2023**.

Teaching

Cornell Tech — *New York, NY*

3x Graduate TA — *Aug 2024 – Dec 2025*

TA'd the graduate-level "Trustworthy AI," "Security and Privacy Concepts in the Wild," and "Privacy in the Digital Age" courses.

University of Washington — Seattle — *Seattle, WA*

4x Undergrad / Grad Machine Learning TA — *Mar 2020 – Dec 2021*

During Spring 2020, Winter 2021, Spring 2021, Autumn 2021:

- Taught undergraduate and graduate students as an undergraduate through 25-person sections and biweekly office hours.
- Designed section materials for entire teaching staff, monitored discussion boards, and graded assignments.

University of Washington — Seattle — *Seattle, WA*

Machine Learning Course Designer — *Jun 2021 – Sep 2021*

Redesigned course materials and infrastructure to improve accessibility and equity in ML education, under **Prof. Sewoong Oh**.

- Revamped problem sets and homework infrastructure to reflect a fast-evolving field and lower barriers to entry.
- Incorporated data context and fairness considerations to help students critically engage with algorithmic bias.
- Developed a centralized grading system and TA codebase for long-term course support.

Service

2025	**Host**, NYC Privacy Day	*New York, NY*
2025	**Strategic Officer**, PhDs at Cornell Tech (PACT)	*New York, NY*
202X	**Reviewer**, ICLR (3x), NeurIPS, COLM, ICML	*Remote*